

KOHLER AI Designer

Core Approach

KOHLER AI Bathroom Designer & Planner (TRACK 1)

01 User Requirements

Room size, budget, style, fixtures, optional floor plan

02 AI Understanding

NLP extraction, vision analysis, conversational copilot

03 KOHLER Product Retrieval

Grounded in real products from the KOHLER catalog

04 Spatial Validation

Clearances, collisions, door swing, zoning

05 Optimization

Space, budget, style, function, sustainability

06 Final Bathroom Design

Synced 2D/3D view with product and BOM info

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KOHLER AI Designer

System Architecture

Each layer passes structured data to the next: AI interprets the request, the intelligence layer grounds it in real KOHLER products, and the engineering layer verifies feasibility before anything is shown.

01

User Input

Dimensions, budget, preferences, natural language, photo or floor plan

02

AI / Vision Layer

Requirement extraction, image and floor-plan analysis, conversational copilot

03

Intelligence Layer

KOHLER catalog retrieval / RAG, product matching, multi-objective optimization

04

Engineering Layer

Geometry engine, collision detection, clearances, door swing, zoning, layout validation

05

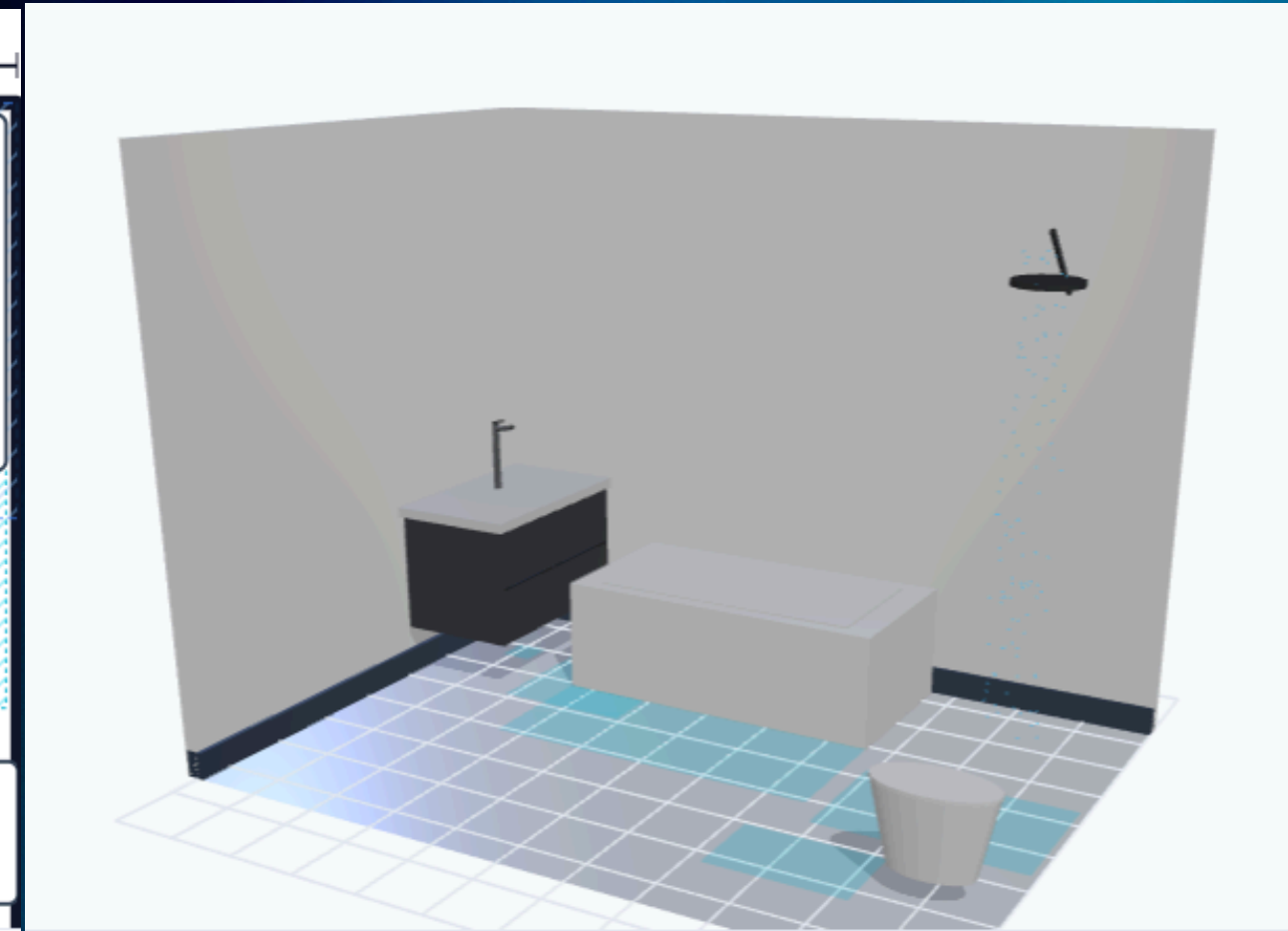
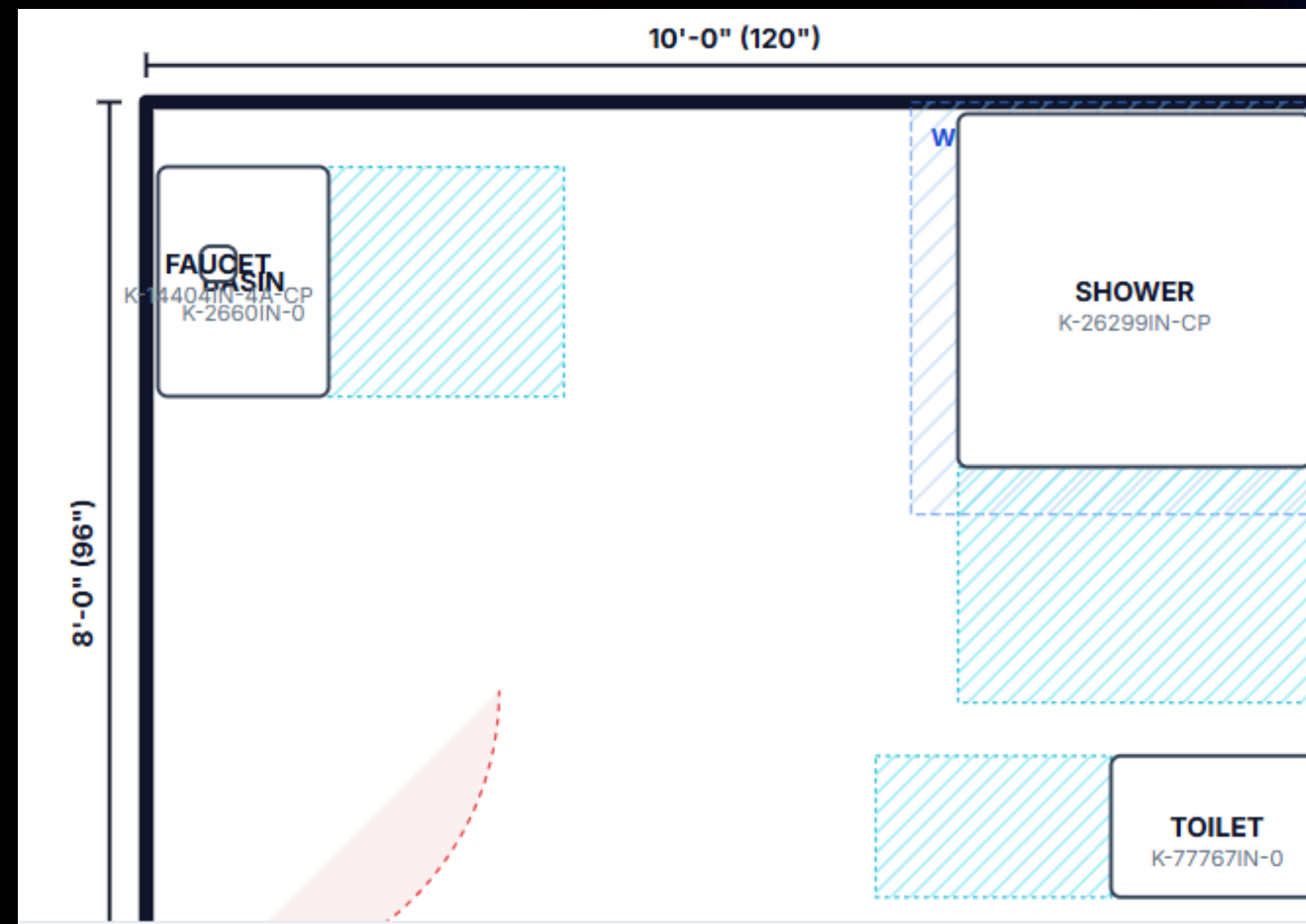
Output Layer

2D floor plan, 3D WebGL studio, sustainability what-if analysis, BOM

AI proposes. Engineering verifies.

KOHLER AI Designer

Tech Stack & Working System



Frontend

React 18, TypeScript, Vite

Backend

FastAPI, Python, Pydantic

3D

Three.js / WebGL

Data

KOHLER product catalog with product / SKU information

AI / Intelligence

NLP requirement extraction, vision analysis, grounded product retrieval / RAG, conversational copilot

Optimization / Engineering

Spatial geometry, collision detection, clearance validation, scoring and alternative generation

System Outputs

2D floor plan, 3D visualization, design alternatives, sustainability what-if analysis, BOM

AI proposes. Engineering verifies.

KOHLER AI Designer

Innovation Pitch

**AI proposes. Engineering
verifies.**

Inspiration → Feasible Design → Product
Selection → Purchase Planning

01

Grounded AI

Recommendations tied to real KOHLER products and SKUs, not imaginary fixtures

02

Feasibility-Aware Design

Clearances, collisions, door swing, and zoning are checked before a layout is shown

03

Multi-Objective Decision Making

Balances aesthetics, space, budget, functionality, and sustainability

04

Design-to-Purchase Continuity

Design connects to product information, pricing, and a procurement-oriented BOM